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The benchmarks in this course are mastery goals that students are expected to attain by the end of the year. To build mastery, students will continue to review and apply earlier grade-level benchmarks and expectations.

Grade Two Mathematics (#5012040)

Version: 2024 - And Beyond (current)

Course Version Note

This version of the course is updated to include additional certification information. No standards, benchmarks or clarifications have been changed.

Course Standards

Visit the specific benchmark webpage to find related instructional resources.

- [MA.2.AR.1.1](#): Solve one- and two-step addition and subtraction real-world problems.

Clarifications:

Clarification 1: Instruction includes understanding the context of the problem, as well as the quantities within the problem.

Clarification 2: Problems include creating real-world situations based on an equation.

Clarification 3: Addition and subtraction are limited to sums up to 100 and related differences. Refer to [Situations Involving Operations with Numbers \(Appendix A\)](#).

- [MA.2.AR.2.1](#): Determine and explain whether equations involving addition and subtraction are true or false.

Clarifications:

Clarification 1: Instruction focuses on understanding of the equal sign.

Clarification 2: Problem types are limited to an equation with three or four terms. The sum or difference can be on either side of the equal sign.

Clarification 3: Addition and subtraction are limited to sums up to 100 and related differences.

Examples:

The equation $27+13=26+14$ can be determined to be true because 26 is one less than 27 and 14 is one more than 13.

- [MA.2.AR.2.2](#): Determine the unknown whole number in an addition or subtraction equation, relating three or four whole numbers, with the unknown in any position.

Clarifications:

Clarification 1: Instruction extends the development of algebraic thinking skills where the symbolic representation of the unknown uses any symbol other than a letter.

Clarification 2: Problems include having the unknown on either side of the equal sign.

Clarification 3: Addition and subtraction are limited to sums up to 100 and related differences. Refer to [Situations Involving Operations with Numbers \(Appendix A\)](#).

Examples:

Determine the unknown in the equation $45 + \square = 23 + 46$.

- [MA.2.AR.3.1](#): Represent an even number using two equal groups or two equal addends. Represent an odd number using two equal groups with one left over or two equal addends plus 1.

Clarifications:

Clarification 1: Instruction focuses on the connection of recognizing

even and odd numbers using skip counting, arrays and patterns in the ones place.

Clarification 2: Addends are limited to whole numbers less than or equal to 12.

Examples:

Example: The number 8 is even because it can be represented as two equal groups of 4 or as the expression $4+4$.

Example: The number 9 is odd because it can be represented as two equal groups with one left over or as the expression $4+4+1$.

- [MA.2.AR.3.2:](#) Use repeated addition to find the total number of objects in a collection of equal groups. Represent the total number of objects using rectangular arrays and equations.

Clarifications:

Clarification 1: Instruction includes making a connection between arrays and repeated addition, which builds a foundation for multiplication.

Clarification 2: The total number of objects is limited to 25.

- [MA.2.DP.1.1:](#) Collect, categorize and represent data using tally marks, tables, pictographs or bar graphs. Use appropriate titles, labels and units.

Clarifications:

Clarification 1: Data displays can be represented both horizontally and vertically. Scales on graphs are limited to ones, fives or tens.

- [MA.2.DP.1.2:](#) Interpret data represented with tally marks, tables, pictographs or bar graphs including solving addition and subtraction problems.

Clarifications:

Clarification 1: Addition and subtraction problems are limited to whole numbers with sums within 100 and related differences.

Clarification 2: Data displays can be represented both horizontally and vertically. Scales on graphs are limited to ones, fives or tens.

- [MA.2.FR.1.1:](#) Partition circles and rectangles into two, three or four equal-sized parts. Name the parts using appropriate language, and describe the whole as two halves, three thirds or four fourths.

Clarifications:

Clarification 1: Within this benchmark, the expectation is not to write the equal-sized parts as a fraction with a numerator and denominator.

Clarification 2: Problems include mathematical and real-world context.

- [MA.2.FR.1.2:](#) Partition rectangles into two, three or four equal-sized parts in two different ways showing that equal-sized parts of the same whole may have different shapes.

Examples:

A square cake can be cut into four equal-sized rectangular pieces or into four equal-sized triangular pieces.

- [MA.2.GR.1.1:](#) Identify and draw two-dimensional figures based on their defining attributes. Figures are limited to triangles, rectangles, squares, pentagons, hexagons and octagons.

Clarifications:

Clarification 1: Within this benchmark, the expectation includes the use of rulers and straight edges.

- [MA.2.GR.1.2:](#) Categorize two-dimensional figures based on the number and length of sides, number of vertices, whether they are closed or not and whether the edges are curved or straight.

Clarifications:

Clarification 1: Instruction focuses on using formal and informal language to describe defining attributes when categorizing.

- [MA.2.GR.1.3:](#) Identify line(s) of symmetry for a two-dimensional figure.

Clarifications:

Clarification 1: Instruction focuses on the connection between partitioning two-dimensional figures and symmetry.

Clarification 2: Problem types include being given an image and determining whether a given line is a line of symmetry or not.

Examples:

Fold a rectangular piece of paper and determine whether the fold is a line of symmetry by matching the two halves exactly.

- [MA.2.GR.2.1](#): Explore perimeter as an attribute of a figure by placing unit segments along the boundary without gaps or overlaps. Find perimeters of rectangles by counting unit segments.

Clarifications:

Clarification 1: Instruction emphasizes the conceptual understanding that perimeter is an attribute that can be measured for a two-dimensional figure.

Clarification 2: Instruction includes real-world objects, such as picture frames or desktops.

- [MA.2.GR.2.2](#): Find the perimeter of a polygon with whole-number side lengths. Polygons are limited to triangles, rectangles, squares and pentagons.

Clarifications:

Clarification 1: Instruction includes the connection to the associative and commutative properties of addition. Refer to [Properties of Operations, Equality and Inequality \(Appendix D\)](#).

Clarification 2: Within this benchmark, the expectation is not to use a formula to find perimeter.

Clarification 3: Instruction includes cases where the side lengths are given or measured to the nearest unit.

Clarification 4: Perimeter cannot exceed 100 units and responses include the appropriate units.

- [MA.2.M.1.1](#): Estimate and measure the length of an object to the nearest inch, foot, yard, centimeter or meter by selecting and using an appropriate tool.

Clarifications:

Clarification 1: Instruction includes seeing rulers and tape measures as number lines.

Clarification 2: Instruction focuses on recognizing that when an object is measured in two different units, fewer of the larger units are required. When comparing measurements of the same object in different units, measurement conversions are not expected.

Clarification 3: When estimating the size of an object, a comparison with an object of known size can be used.

- [MA.2.M.1.2](#): Measure the lengths of two objects using the same unit and determine the difference between their measurements.

Clarifications:

Clarification 1: Within this benchmark, the expectation is to measure objects to the nearest inch, foot, yard, centimeter or meter.

- [MA.2.M.1.3](#): Solve one- and two-step real-world measurement problems involving addition and subtraction of lengths given in the same units.

Clarifications:

Clarification 1: Addition and subtraction problems are limited to sums within 100 and related differences.

Examples:

Jeff and Larry are making a rope swing. Jeff has a rope that is 48 inches long. Larry's rope is 9 inches shorter than Jeff's. How much rope do they have together to make the rope swing?

- [MA.2.M.2.1](#): Using analog and digital clocks, tell and write time to the nearest five minutes using a.m. and p.m. appropriately. Express portions of an hour using the fractional terms half an hour, half past, quarter of an hour, quarter after and quarter til.

Clarifications:

Clarification 1: Instruction includes the connection to partitioning of circles and to the number line.

Clarification 2: Within this benchmark, the expectation is not to understand military time.

- [MA.2.M.2.2](#): Solve one- and two-step addition and subtraction real-world problems involving either dollar bills within \$100 or coins within 100¢ using \$ and ¢ symbols appropriately.

Clarifications:

Clarification 1: Within this benchmark, the expectation is not to use decimal values.

Clarification 2: Addition and subtraction problems are limited to sums within 100 and related differences. Refer to [Situations Involving Operations with Numbers \(Appendix A\)](#).

- [MA.2.NSO.1.1](#): Read and write numbers from 0 to 1,000 using standard form, expanded form and word form.

Examples:

Example: The number four hundred thirteen written in standard form is 413 and in expanded form is $400+10+3$.

Example: The number seven hundred nine written in standard form is 709 and in expanded form is $700+9$.

- [MA.2.NSO.1.2:](#) Compose and decompose three-digit numbers in multiple ways using hundreds, tens and ones. Demonstrate each composition or decomposition with objects, drawings and expressions or equations.

Examples:

The number 241 can be expressed as 2 hundreds + 4 tens + 1 one or as 24 tens + 1 one or as 241 ones.

- [MA.2.NSO.1.3:](#) Plot, order and compare whole numbers up to 1,000.

Clarifications:

Clarification 1: When comparing numbers, instruction includes using a number line and using place values of the hundreds, tens and ones digits.

Clarification 2: Within this benchmark, the expectation is to use terms (e.g., less than, greater than, between or equal to) and symbols ($<$, $>$ or $=$).

Examples:

The numbers 424, 178 and 475 can be arranged in ascending order as 178, 424 and 475.

- [MA.2.NSO.1.4:](#) Round whole numbers from 0 to 100 to the nearest 10.

Clarifications:

Clarification 1: Within the benchmark, the expectation is to understand that rounding is a process that produces a number with a similar value that is less precise but easier to use.

Examples:

The number 65 is rounded to 70 when rounded to the nearest 10.

- [MA.2.NSO.2.1:](#) Recall addition facts with sums to 20 and related subtraction facts with automaticity.
- [MA.2.NSO.2.2:](#) Identify the number that is ten more, ten less, one hundred more and one hundred less than a given three-digit number.

Examples:

The number 236 is one hundred more than 136 because both numbers have the same digit in the ones and tens place, but differ in the hundreds place by one.

- [MA.2.NSO.2.3](#): Add two whole numbers with sums up to 100 with procedural reliability. Subtract a whole number from a whole number, each no larger than 100, with procedural reliability.

Clarifications:

Clarification 1: Instruction focuses on helping a student choose a method they can use reliably.

Examples:

Example: The sum $41+23$ can be found by using a number line and “jumping up” by two tens and then by three ones to “land” at 64.

Example: The difference $87-25$ can be found by subtracting 20 from 80 to get 60 and then 5 from 7 to get 2. Then add 60 and 2 to obtain 62.

- [MA.2.NSO.2.4](#): Explore the addition of two whole numbers with sums up to 1,000. Explore the subtraction of a whole number from a whole number, each no larger than 1,000.

Clarifications:

Clarification 1: Instruction includes the use of manipulatives, number lines, drawings or properties of operations or place value.

Clarification 2: Instruction focuses on composing and decomposing ones, tens and hundreds when needed.

Examples:

Example: The difference $612-17$ can be found by rewriting it as $612-12-5$ which is equivalent to $600-5$ which is equivalent to 595.

Example: The difference $1,000-17$ can be found by using a number line and making a “jump” of 10 from 1,000 to 990 and then 7 “jumps” of 1 to 983.

- [MA.K12.MTR.1.1](#): **Actively participate in effortful learning both individually and collectively.**

Mathematicians who participate in effortful learning both individually and with others:

- Analyze the problem in a way that makes sense given the task.
- Ask questions that will help with solving the task.
- Build perseverance by modifying methods as needed while solving a challenging task.
- Stay engaged and maintain a positive mindset when working to solve tasks.
- Help and support each other when attempting a new method or approach.

Clarifications:

Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of growth mindset learners.
- Foster perseverance in students by choosing tasks that are challenging.
- Develop students' ability to analyze and problem solve.
- Recognize students' effort when solving challenging problems.

• **MA.K12.MTR.2.1: Demonstrate understanding by representing problems in multiple ways.**

Mathematicians who demonstrate understanding by representing problems in multiple ways:

- Build understanding through modeling and using manipulatives.
- Represent solutions to problems in multiple ways using objects, drawings, tables, graphs and equations.
- Progress from modeling problems with objects and drawings to using algorithms and equations.
- Express connections between concepts and representations.
- Choose a representation based on the given context or purpose.

Clarifications:

Teachers who encourage students to demonstrate understanding by representing problems in multiple ways:

- Help students make connections between concepts and representations.
- Provide opportunities for students to use manipulatives when investigating concepts.
- Guide students from concrete to pictorial to abstract representations as understanding progresses.
- Show students that various representations can have different purposes and can be useful in different situations.
- **MA.K12.MTR.3.1: Complete tasks with mathematical fluency.**

Mathematicians who complete tasks with mathematical fluency:

- Select efficient and appropriate methods for solving problems within the given context.
- Maintain flexibility and accuracy while performing procedures and mental calculations.
- Complete tasks accurately and with confidence.
- Adapt procedures to apply them to a new context.
- Use feedback to improve efficiency when performing calculations.

Clarifications:

Teachers who encourage students to complete tasks with mathematical fluency:

- Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
- Offer multiple opportunities for students to practice efficient and generalizable methods.
- Provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.

- **MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others.**

Mathematicians who engage in discussions that reflect on the mathematical thinking of self and others:

- Communicate mathematical ideas, vocabulary and methods effectively.
- Analyze the mathematical thinking of others.

- Compare the efficiency of a method to those expressed by others.
- Recognize errors and suggest how to correctly solve the task.
- Justify results by explaining methods and processes.
- Construct possible arguments based on evidence.

Clarifications:

Teachers who encourage students to engage in discussions that reflect on the mathematical thinking of self and others:

- Establish a culture in which students ask questions of the teacher and their peers, and error is an opportunity for learning.
- Create opportunities for students to discuss their thinking with peers.
- Select, sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
- Develop students' ability to justify methods and compare their responses to the responses of their peers.

• **MA.K12.MTR.5.1: Use patterns and structure to help understand and connect mathematical concepts.**

Mathematicians who use patterns and structure to help understand and connect mathematical concepts:

- Focus on relevant details within a problem.
- Create plans and procedures to logically order events, steps or ideas to solve problems.
- Decompose a complex problem into manageable parts.
- Relate previously learned concepts to new concepts.
- Look for similarities among problems.
- Connect solutions of problems to more complicated large-scale situations.

Clarifications:

Teachers who encourage students to use patterns and structure to help understand and connect mathematical concepts:

- Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts.
- Support students to develop generalizations based on the similarities found among problems.

- Provide opportunities for students to create plans and procedures to solve problems.
- Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.

- **MA.K12.MTR.6.1: Assess the reasonableness of solutions.**

Mathematicians who assess the reasonableness of solutions:

- Estimate to discover possible solutions.
- Use benchmark quantities to determine if a solution makes sense.
- Check calculations when solving problems.
- Verify possible solutions by explaining the methods used.
- Evaluate results based on the given context.

Clarifications:

Teachers who encourage students to assess the reasonableness of solutions:

- Have students estimate or predict solutions prior to solving.
- Prompt students to continually ask, "Does this solution make sense? How do you know?"
- Reinforce that students check their work as they progress within and after a task.
- Strengthen students' ability to verify solutions through justifications.

- **MA.K12.MTR.7.1: Apply mathematics to real-world contexts.**

Mathematicians who apply mathematics to real-world contexts:

- Connect mathematical concepts to everyday experiences.
- Use models and methods to understand, represent and solve problems.
- Perform investigations to gather data or determine if a method is appropriate.
- Redesign models and methods to improve accuracy or efficiency.

Clarifications:

Teachers who encourage students to apply mathematics to real-world contexts:

- Provide opportunities for students to create models, both concrete and abstract, and perform investigations.
- Challenge students to question the accuracy of their models and methods.
- Support students as they validate conclusions by comparing them to the given situation.
- Indicate how various concepts can be applied to other disciplines.
- [ELA.K12.EE.1.1](#): Cite evidence to explain and justify reasoning.

Clarifications:

K-1 Students include textual evidence in their oral communication with guidance and support from adults. The evidence can consist of details from the text without naming the text. During 1st grade, students learn how to incorporate the evidence in their writing.

2-3 Students include relevant textual evidence in their written and oral communication. Students should name the text when they refer to it. In 3rd grade, students should use a combination of direct and indirect citations.

4-5 Students continue with previous skills and reference comments made by speakers and peers. Students cite texts that they've directly quoted, paraphrased, or used for information. When writing, students will use the form of citation dictated by the instructor or the style guide referenced by the instructor.

6-8 Students continue with previous skills and use a style guide to create a proper citation.

9-12 Students continue with previous skills and should be aware of existing style guides and the ways in which they differ.

- [ELA.K12.EE.2.1](#): Read and comprehend grade-level complex texts proficiently.

Clarifications:

See [Text Complexity](#) for grade-level complexity bands and a text complexity rubric.

- [ELA.K12.EE.3.1](#): Make inferences to support comprehension.

Clarifications:

Students will make inferences before the words infer or inference are introduced. Kindergarten students will answer questions like "Why is the girl smiling?" or make predictions about what will

happen based on the title page. Students will use the terms and apply them in 2nd grade and beyond.

- [ELA.K12.EE.4.1](#): Use appropriate collaborative techniques and active listening skills when engaging in discussions in a variety of situations.

Clarifications:

In kindergarten, students learn to listen to one another respectfully.

In grades 1-2, students build upon these skills by justifying what they are thinking. For example: “I think _____ because _____.” The collaborative conversations are becoming academic conversations.

In grades 3-12, students engage in academic conversations discussing claims and justifying their reasoning, refining and applying skills. Students build on ideas, propel the conversation, and support claims and counterclaims with evidence.

- [ELA.K12.EE.5.1](#): Use the accepted rules governing a specific format to create quality work.

Clarifications:

Students will incorporate skills learned into work products to produce quality work. For students to incorporate these skills appropriately, they must receive instruction. A 3rd grade student creating a poster board display must have instruction in how to effectively present information to do quality work.

- [ELA.K12.EE.6.1](#): Use appropriate voice and tone when speaking or writing.

Clarifications:

In kindergarten and 1st grade, students learn the difference between formal and informal language. For example, the way we talk to our friends differs from the way we speak to adults. In 2nd grade and beyond, students practice appropriate social and academic language to discuss texts.

- [ELD.K12.ELL.MA.1](#): English language learners communicate information, ideas and concepts necessary for academic success in the content area of Mathematics.
- [ELD.K12.ELL.SI.1](#): English language learners communicate for social and instructional purposes within the school setting.

Course Information

General Notes

In grade 2, instructional time will emphasize four areas: (1) extending understanding of place value in three-digit numbers; (2) building fluency and algebraic reasoning with addition and subtraction; (3) extending understanding of measurement of objects, time and the perimeter of geometric figures and (4) developing spatial reasoning with number representations and two-dimensional figures.

Curricular content for all subjects must integrate critical-thinking, problem-solving, and workforce-literacy skills; communication, reading, and writing skills; mathematics skills; collaboration skills; contextual and applied-learning skills; technology-literacy skills; information and media-literacy skills; and civic-engagement skills.

English Language Development ELD Standards Special Notes Section:

Teachers are required to provide listening, speaking, reading and writing instruction that allows English language learners (ELL) to communicate information, ideas and concepts for academic success in the content area of Mathematics. For the given level of English language proficiency and with visual, graphic, or interactive support, students will interact with grade level words, expressions, sentences and discourse to process or produce language necessary for academic success. The ELD standard should specify a relevant content area concept or topic of study chosen by curriculum developers and teachers which maximizes an ELL's need for communication and social skills. To access an ELL supporting document which delineates performance definitions and descriptors, please click on the following link:

https://cpalmsmediaproduct.blob.core.windows.net/uploads/docs/standard_s/eld/ma.pdf

General Information

Course Number: 5012040

Course Path: **Section:** Grades PreK to 12 Education Courses >

Grade Group: Grades PreK to 5 Education Courses > **Subject:**

Mathematics > **SubSubject:** General Mathematics >

Abbreviated Title: GRADE TWO MATH

Course Length: Year (Y)

Course Type: Core Academic Course

Course Level: 2

Course Status: State Board Approved

Grade Level(s): K,1,2,3,4,5

Educator Certifications

One of these educator certification options is required to teach this course.

- [Prekindergarten/Primary Education \(Age 3 through Grade 3\)](#)
- [Elementary Education \(Elementary Grades 1-6\)](#)
- [Primary Education \(K-3 - No Longer Issued\)](#)
- [Mathematics \(Elementary Grades 1-6\)](#)
- [Elementary Education \(Grades K-6\)](#)
- [Classical Education - Restricted \(Elementary and Secondary Grades K-12\)](#)

Section 1012.55(5), F.S., authorizes the issuance of a classical education teaching certificate, upon the request of a classical school, to any applicant who fulfills the requirements of s. 1012.56(2)(a)-(f) and (11), F.S., and Rule 6A-4.004, F.A.C. Classical schools must meet the requirements outlined in s. 1012.55(5), F.S., and be listed in the [FLDOE Master School ID](#) database, to request a restricted classical education teaching certificate on behalf of an applicant.